

1. Describe the structure and operation of the DES algorithm, including its key size and block size.

Structure and Operation of the DES Algorithm

The Data Encryption Standard (DES) is an **iterated block cipher** based on the **Feistel Cipher** structure. It was the U.S. government standard developed in the 1970s.

1. Key and Block Size

- **Block Size:** DES processes data in fixed-sized blocks of **64 bits**. This means both the input (plaintext) and output (ciphertext) are 64 bits long.
 - **Key Size:** The algorithm uses a **56-bit key**. Although the initial key input is often 64 bits, 8 bits are used for parity, leaving 56 bits for the actual encryption process.
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2. Structure (The Feistel Network)

DES follows the Feistel structure, which means:

- The 64-bit plaintext is split into two halves: **Left (L_0)** and **Right (R_0)**, each being 32 bits.
 - It performs **16 rounds** of identical operations to transform the plaintext into ciphertext.
 - The decryption process is essentially the same as encryption, but the round subkeys are used in reverse order.
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3. Detailed Operation (Step-by-Step)

1. **Initial Permutation (IP):** Before the first round, the 64-bit block undergoes a fixed bit-shuffling process.
2. **The 16 Rounds:** In each round ($i=1$ to 16), the following steps occur: **(Etar jonno Feistel Cipher: Encryption and Decryption er process ta ektu dekhe then eta lekhar try korbo tahole sohoj hobe but oi Algorithm tao pore nebo)**
 - **Expansion:** The 32-bit Right half (R_{i-1}) is expanded to **48 bits** using an Expansion Permutation.
 - **Key Mixing:** The expanded 48 bits are XORed with a **48-bit subkey (K_i)** generated from the main 56-bit key.
 - **Substitution (S-boxes):** The 48-bit result is fed into **8 S-boxes**. Each S-box maps 6 input bits to 4 output bits, resulting in a **32-bit** output. This is the most critical part of DES security as it provides non-linearity.
 - **P-box Permutation:** The 32-bit output from the S-boxes is shuffled again through a fixed permutation.
 - **XOR and Swap:** The result is XORed with the Left half (L_{i-1}). Finally, the Left and Right halves are swapped for the next round.

3. **Final Permutation (IP⁻¹):** After the 16th round, the halves are swapped back, and a final permutation (the inverse of the IP) is applied to produce the final **64-bit ciphertext**.

Feature	Specification
Algorithm Type	Feistel Cipher (Symmetric)
Block Length	64 bits
Key Length	56 bits
Number of Rounds	16
Subkey Size	48 bits (per round)
Non-linear Element	8 S-boxes

2. What role do S-boxes play in block ciphers like DES?

The role of S-boxes:

- **Non-linearity:** They are the **only** non-linear part of DES. Without them, the encryption would be a simple math problem that is easy to crack.
- **Confusion:** They hide the relationship between the plaintext and the ciphertext, making it impossible for attackers to find a pattern.
- **Core Security:** The overall security of DES depends almost entirely on the S-boxes.
- **Bit Mapping:** Each S-box converts **6 input bits** into **4 output bits**.

In short: S-boxes provide the "strength" of the encryption by making it mathematically complex and secure.

S-box-er vumika (Bangla-y):

- **Non-linearity (অরৈখিকতা):** এগুলি DES-এর একমাত্র নন-লিনিয়ার অংশ। এগুলি ছাড়া এনক্রিপশনটি একটি সাধারণ গাণিতিক সমস্যায় পরিণত হতো, যা খুব সহজেই সমাধান (crack) করা সম্ভব হতো।
- **Confusion (বিভ্রান্তি):** এগুলি প্লেইনটেক্সট এবং সাইফারটেক্সটের মধ্যে সম্পর্ককে লুকিয়ে রাখে, যার ফলে আক্রমণকারীদের পক্ষে কোনো প্যাটার্ন খুঁজে পাওয়া অসম্ভব হয়ে পড়ে।

- **Core Security (মূল নিরাপত্তা):** DES-এর সামগ্রিক নিরাপত্তা প্রায় সম্পূর্ণভাবে এই S-box-গুলোর ওপর নির্ভর করে।
- **Bit Mapping (বিট ম্যাপিং):** প্রতিটি S-box ৬-বিট ইনপুটকে ৪-বিট আউটপুটে রূপান্তর করে।

সংক্ষেপে: S-box এনক্রিপশনকে গাণিতিকভাবে জটিল এবং সুরক্ষিত করার মাধ্যমে এর আসল "শক্তি" প্রদান করে।

4. Why DES Encrypt-Decrypt-Encrypt with 2 keys?

Based on your dataset, the **Encrypt-Decrypt-Encrypt (EDE)** sequence with two keys (K_1 and K_2) is used in **Triple DES (3DES)** for two main reasons:

- **Backward Compatibility:** By using the EDE sequence, if you set both keys to be the same ($K_1 = K_2$), the operation becomes identical to standard single DES. This allowed older systems to communicate with newer ones easily.
 - Formula: $E(D(E(P, K), K), K) = E(P, K)$
- **Increased Key Length:** Standard DES has a 56-bit key, which is too small and easy to crack. By using two different keys (K_1 and K_2), the effective key length increases to **112 bits**, making exhaustive key searches (brute force) practically impossible.

3DES-এ ২-টি কী (key) ব্যবহার করে **Encrypt-Decrypt-Encrypt (EDE)** সিকোয়েন্স কেন ব্যবহার করা হয়, তার কারণগুলো নিচে বাংলায় দেওয়া হলো:

- **ব্যাকওয়ার্ড কম্প্যাটিবিলিটি (Backward Compatibility):** EDE সিকোয়েন্স ব্যবহার করার ফলে, যদি আপনি দুটি কী-ই একই রাখেন ($K_1 = K_2$), তবে পুরো অপারেশনটি সাধারণ সিঙ্গেল DES-এর মতো কাজ করবে। এর ফলে পুরনো সিস্টেমগুলো সহজেই নতুন সিস্টেমের সাথে যোগাযোগ করতে পারতো।
 - সূত্র: $E(D(E(P, K), K), K) = E(P, K)$
- **কী-এর দৈর্ঘ্য বৃদ্ধি (Increased Key Length):** সাধারণ DES-এর কী মাত্র ৫৬-বিটের, যা বর্তমান সময়ে ক্র্যাক করা বা ভেঙে ফেলা খুব সহজ। দুটি আলাদা কী (K_1 এবং K_2) ব্যবহারের ফলে কার্যকর কী-এর দৈর্ঘ্য বেড়ে **১১২-বিট** হয়। এর ফলে এক্সজস্টিভ কী সার্চ (Exhaustive Key Search) বা ব্রুট ফোর্স অ্যাটাকের মাধ্যমে পাসওয়ার্ড বের করা প্রায় অসম্ভব হয়ে পড়ে।

In DES cipher.

- a. How many bits in each plaintext block?
- b. How many bits in each ciphertext block?
- c. How many bits in the key?

- d. How many bits in each subkey?
- e. How many rounds?
- f. How many S-boxes?
- g. An S-box requires how many bits of input?
- h. An S-box generates how many bits of output?

Based on the "DES Numerology" section of your dataset, here are the direct answers to your questions:

- **a. How many bits in each plaintext block?**

64 bits

- **b. How many bits in each ciphertext block?**

64 bits

- **c. How many bits in the key?**

56 bits (often referred to as the effective key length)

- **d. How many bits in each subkey?**

48 bits

- **e. How many rounds?**

16 rounds

- **f. How many S-boxes?**

8 S-boxes

- **g. An S-box requires how many bits of input?**

6 bits

- **h. An S-box generates how many bits of output?**

4 bits

5. What improvements does AES offer over DES? List its key features and block size.

Improvements of AES over DES:

- **Variable Key Lengths:** Unlike DES, which uses a fixed 56-bit key, AES allows for much longer keys (128, 192, or 256 bits), making it exponentially harder to break via brute-force attacks.
- **Larger Block Size:** AES uses a 128-bit block size, whereas DES uses a 64-bit block size. A larger block size is more secure against certain types of cryptanalytic attacks.
- **Modern Structure:** AES is based on a **Substitution-Permutation Network (SPN)** consisting of three specific layers (Nonlinear, Linear Mixing, and Key Addition), whereas DES is based on the Feistel network.
- **Efficiency:** AES is designed to be efficient in both software and hardware implementations.

Key Features of AES

According to the "AES: Executive Summary" in your notes, the core features are:

- **Block Size: 128 bits** (Note: The Rijndael algorithm supports other sizes, but AES is standardized at 128).
- **Key Lengths: 128, 192, or 256 bits** (independent of the block size).
- **Number of Rounds: 10 to 14 rounds**, depending on the length of the key.
- **Round Functions:** Each round (except the final one) consists of four specific functions:
 1. **ByteSub:** A nonlinear layer (the S-box) used for substitution.
 2. **ShiftRow:** A linear mixing layer that cyclically shifts the rows of the 4x4 byte array.
 3. **MixColumn:** A nonlinear layer that provides diffusion by mixing the data within each column (implemented as a lookup table).
 4. **AddRoundKey:** A key addition layer where a subkey is XORed with the block.

Summary Table

Feature	AES (Standard)	DES (Reference)
Block Size	128 bits	64 bits
Key Length	128, 192, or 256 bits	56 bits
Algorithm Structure	Substitution-Permutation Network	Feistel Network
Security	High (Current Standard)	Low (Considered insecure for modern use)

6. Suppose you have the following S-block of a DES model.

Column →	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Row 0	1 4	4	1 3	1	2	1 5	1 1	8	3	1 0	6	1 2	5	9	0	7
Row 1	0	1 5	7	4	1 4	2	1 3	1	1 0	6	1 2	1 1	9	5	3	8
Row 2	4	1	1 4	8	1 3	6	2	1 1	1 5	1 2	9	7	3	1 0	5	0
Row 3	1 5	1 2	8	2	4	9	1	7	5	1 1	3	1 4	1 0	0	6	1 3

Now, for the input sequence 101101, answer the following.

i. Which row contains the output value?

Answer: 3

ii. Which column contains the output value?

Answer: 6

iii. What will be the value of the output in decimal?

Answer: 1

iv. What will be the value of the output in binary?

An S-box output is always 4 bits. The decimal value 1 converted to a 4-bit binary sequence is **0001**.

- **Answer:** 0001

7. Describe the four functions used in an AES encryption round.

Four Functions in an AES Round

- **ByteSub:** Replace characters (Substitution).

Acts as the **S-box** for AES.

What it does: Swaps characters for new ones.

Purpose: To make the relationship between the original text and the scrambled text very confusing.

- **ShiftRow:** Slide rows horizontally (Shuffling).
 - **1st Row:** Stays where it is.
 - **2nd Row:** Slides **1 position** to the left.
 - **3rd Row:** Slides **2 positions** to the left.
 - **4th Row:** Slides **3 positions** to the left.

- **Purpose:** Moves the data around horizontally so everything is out of its original place.
- **MixColumn:** Blend columns vertically (Mixing).
 - linear operation applied to **each column** of the 4x4 array.
 - this is implemented as a **large lookup table** to ensure the operation is fast and efficient.

Purpose: Change in one byte spreads across multiple bytes.

- **AddRoundKey:** Apply the secret password (Locking).

What it does: Uses the **XOR** math operation to "add" the key to the data.

Purpose: Making the output unpredictable without the key. This is the actual "lock" on the data. Without the correct secret key, no one can reverse the other steps to read the message.

8. AES consists of four functions in three layers. o a. Which of the four functions are primarily for confusion and which are primarily for diffusion? Justify your answer.

AES uses four functions: **ByteSub**, **ShiftRow**, **MixColumn**, **AddRoundKey**.

Confusion (make the data hard to understand):

- **ByteSub:** Replace characters (Substitution).

Acts as the **S-box** for AES.

What it does: Swaps characters for new ones.

Purpose: To make the relationship between the original text and the scrambled text very confusing.

- **AddRoundKey:** Apply the secret password (Locking).

What it does: Uses the **XOR** math operation to "add" the key to the data.

Purpose: Making the output unpredictable without the key.

Diffusion (spread the data changes):

ShiftRow:

→ It shifts rows of the data.

→ This moves bytes to different positions.

✓ **Provides diffusion**

MixColumn: Blend columns vertically (Mixing).

- linear operation applied to **each column** of the 4x4 array.
- this is implemented as a **large lookup table** to ensure the operation is fast and efficient.

Purpose: Change in one byte spreads across multiple bytes. **Provides Strong diffusion**

9. Compare ECB (Electronic Codebook) and CBC (Cipher Block Chaining) modes of operation. Highlight their advantages and disadvantages.

◆ ECB (Electronic Codebook)

How it works:

- Each block is encrypted **independently**
- Same plaintext block → same ciphertext block

Advantages:

- Very **simple**
- Easy to implement

Disadvantages:

- ✗ Not secure
- ✗ Same data gives same result (pattern visible)
- ✗ Vulnerable to **cut-and-paste attack**

◆ CBC (Cipher Block Chaining)

How it works:

- Each block depends on the **previous ciphertext block**
- Uses a random **IV (Initialization Vector)**

Advantages:

- ✓ More **secure than ECB**
- ✓ Same plaintext gives **different ciphertext**
- ✓ Hides patterns well
- It can automatically recover from transmission errors after a few blocks.

Disadvantages:

- ✗ Error in one block affects next block
- ✗ Slightly more complex
- ✗ Needs IV (must be random)

Feature	ECB Mode	CBC Mode
Block Independence	Each block is independent.	Blocks are chained together.
Pattern Protection	None; same plaintext = same ciphertext.	Good; same plaintext = different ciphertext.
Security Level	Low (Bad idea for most uses).	High (More secure than ECB).
Key Requirement	Uses a secret key.	Uses a secret key + a random IV.

10. What is the purpose of an initialization vector (IV) in CBC mode?

The IV (Initialization Vector) is used to start CBC encryption with randomness, so the same plaintext produces different ciphertext.

Purpose of IV in CBC mode:

- **Start the chain:** A random value used for the first block.
- **Hide patterns:** Same plaintext gives different ciphertext.
- **Add randomness:** Not secret, but must be random for security.

11. Explain the concept of a Message Authentication Code (MAC). How does it ensure data integrity?

MAC (Message Authentication Code):

A MAC is a small value made from a message using a secret key to check if data is changed.

How it works:

- Sender creates MAC using the message and secret key
- Sender sends message + MAC
- Receiver recalculates MAC
- If both MACs match → data is safe
- If not → data is changed

How it ensures data integrity:

- The receiver **recomputes the MAC** using the same key.
- **If new MAC = received MAC**
 - The receiver's calculated MAC matches the sender's MAC
 - ✓ The data has not been changed (data is intact)
- **If not equal**
 - The MACs are different
 - ✗ The data has been modified during transmission

☞ Simple idea:

Same MAC = safe data

Different MAC = tampered data

A digital signature provides for data integrity and a MAC provides for data integrity. Why does a signature also provides for non-repudiation while a MAC does not?

A **digital signature** provides non-repudiation because it uses **public-key cryptography**:

- The sender signs the message using their **private key**.
 - Anyone can verify it using the sender's **public key**.
- ☞ So only the sender could have created it, and they cannot deny it later.

A **MAC does not provide non-repudiation** because:

- It uses a **shared secret key** between sender and receiver.
 - Both sender and receiver can generate the same MAC.
- ☞ So you cannot prove who actually created it, since both know the key.

Simple idea:

- **Digital signature:** proves *who sent it*
- **MAC:** only proves *data is not changed*, not *who sent it*